

## **Lake Bottom Sediment Coring from Schirmacher Oasis During 24<sup>th</sup> Indian Antarctic Expedition**

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### **ABSTRACT**

Lake sediment cores retrieved from pristine environment of Antarctica can provide a high resolution palaeoclimatic record having a significant bearing on the global climate change. The sedimentary layers at the lake bottom contain chronologically ordered information. Studies on long term climatic changes in Antarctica have been focused on ice core and marine records which have provided insights about atmospheric / marine couplings during last glacial / interglacial cycle. Owing to the lack of large ice-free areas and the presence of permafrost, terrestrial records have played a less significant role in documenting Holocene climatic changes.

The Schirmacher Oasis is interspersed with a number of freshwater lakes offering a great potential of systematic multi-disciplinary studies. The expertise of retrieving in-situ lake bottom sediment cores using indigenously designed and developed coring device by Antarctica Division has yielded significant results. Thus, a systematic attempt of Lake Bottom Sediment Coring was made during the 24<sup>th</sup> Indian Antarctic Expedition mainly during the wintering over period.

### **INTRODUCTION**

Limnological studies from Antarctic Peninsula, McMurdo Dry Valley and East Antarctic lake-rich Oases: Vestfold Hills, Larsemann Hills, Bunger Hills, Rauer Islands and Windmill Islands have already contributed significantly in the various aspects of lacustrine systems. Earlier attempts of retrieving Lake Bottom sediment cores from Schirmacher Oasis succeeded in getting a few cores of about 50 to 70 cm lengths. Studies on these cores led to an inference indicating a three fold climatic oscillation (warm-dry, warm-humid and warm more humid) since last 8,000 BP (Bera and Khandelwal, 2003; Bera, 2004; Bera, 2006) besides identification of the first record of some Arcellaceans (Mathur et al., 2006) which are very important ecological indicators.

Allowing the scope of expansion and improvement of scientific information based on lake bottom sediment cores, a total number of 39 lake bottom sediment cores (cumulative core length - 43.50m) have been retrieved using indigenously designed and fabricated sediment coring device during the austral winter of 24<sup>th</sup> Indian Scientific Expedition to Antarctica (2004 – 06). From each coring site, entire sedimentation column from top to bedrock have been raised with some cores of more than 200 cm lengths (longest core being 221 cm) that are likely to provide data beyond LGM. A comprehensive multidisciplinary investigation including sedimentological, geochemical, biological, palynological, micropalaeontological and mineral magnetic susceptibility studies have been envisaged (Asthana *et al.*, 2006) along with isotopic and thermo luminescence (TL) dating to elucidate various parameters. Synthesis of all these data would be helpful in reconstructing the palaeoclimatic and palaeoenvironmental regimes of East Antarctica that will have a bearing on the global climatic changes in recent geologic past.

#### **Lacustrine Environment at Schirmacher Oasis, Sediment Characteristics and Scope of Investigation**

The “roche moutonnee” landscape of Schirmacher Oasis is characterized by numerous lakes which are mostly ice-scoured basins and troughs (Beg *et al.*, 2006) produced by moving ice streams/ice sheets subsequent to the Last Glacial Maximum. Sediments derived from the ice masses are transported and deposited by melt water and these sedimentary layers at the lake bottom contain chronologically ordered information regarding the topography and environmental history of the region.

The glacial and periglacial environments encompassing lakes of varying dimensions are characterized by specific physical, chemical and biological conditions and the major glacial activities involve both erosion and sedimentation. Lacustrine systems are very sensitive and readily affected by even minor tectonic or climatic changes, interacting tectonics, drainage evolution and sediment supply that get preserved in their depositional records.

All major lakes of Schirmacher Oasis are localized along the glacial valleys (Ravindra *et al.*, 2002; Beg *et al.*, 2006). The older (first generation) valleys are separated by high grounds with typical development of Roche Moutonnees; whereas steep and clean cut walls define the younger ones. In each individual valley, the glaciers, directly feed the southernmost pro-

glacial lakes, and towards north, the successive lakes are partly fed by the seasonal snow and partly through a subsurface flow regime manifested as patterned ground or by elementary development of surface channels. These subsurface as well as surface flow regimes eventually drain the entire valley in north on to the ice shelf, which in turn is marked by huge epi-shelf lakes.

From a sedimentological point of view, the most characteristic feature of the glacial environment is deposition of debris supplied by ice-mass – i.e. settling of material from melting ice in the ablation zone (Asthana and Chaturvedi, 1998). Then, the topographic undulations and irregularities form pools / lakes of variable sizes are characterized by almost stagnant water in which the finest sediment particles settle under typical lacustrine environment. The sediments formed in such lakes are characteristically glacio-lacustrine deposits – which are derived from melting ice around the vicinity of lakes, polar ice-sheet – south of Schirmacher Oasis and ephemeral melt water channels embouching in the lakes along with debris-bearing strong polar winds. Intense action of aeolian sedimentation in the present area remains a conspicuous feature. Wind action coupled with snow storms also erode and transport particles larger than sand size, thus giving rise to coarse niveo-aeolian fractions occurring within the typical lacustrine sediment systems.

Most of the glacio lacustrine sediments in the area are relatively fine grained because the water column generally remains stagnant or has a low flow velocity, so that even the finest particles get settled. Factors responsible for the final depositional process are settling out of suspension (wind blown materials also) by the surficial currents, rolling and traction populations through insignificant bottom currents and little contribution from the loosely held thin veneer of glacial-debris occurring on the sub-aerial or sub-aqueous adjoining slopes.

The granulometric characteristics commonly shows relatively coarse lake margin sediments and fine grained bottom sets; the latter frequently show thinly laminated varves: graded layers that originate from seasonal settling due to glacial melt water in the spring seasons. The sediment cores contain mostly faintly laminated silts or sandy mud with frequent pebble-size rock fragments, which is typical of deposits found in water bodies covered by an ice sheet.

Climatic fluctuations up to the past 32,655 years have been inferred from a lake bottom sediment core from L – 49, Priyadarshini Lake

(Achyuthan *et al.*, 2008). The rate of sedimentation and algal matt growth in a core has been very low varying from 0.005mm per year between 30,640 yr BP to 21,685 yr BP and 0.015mm per year between 32,655 yr BP to 30,640 yr BP (warm period). Another core represents about 0.09mm / yr rate of sedimentation between 29,920 and 28,890 yr BP indicating a wet period. It is noted that during the overall warm period between the Mid MIS 2 (25,000 yrs BP) and beginning of MIS 3 (28,000 onwards), there have been brief periods of warmer and wetter interludes.

The glacio-lacustrine sedimentary processes are characterized by direct deposition from glacier ice at the ice-margin, fluvial deposition from glacial melt water streams feeding the lake system, sedimentation from suspension of material in lake water, sedimentation from gravity-driven flows from the margin of the sub-glacial region where sediment transport is at a maximum, deposition of sediment particles by strong polar winds and three relatively minor non glacial processes, viz. 1. Lake-shore sedimentation, 2. Biogenic sedimentation and 3. Evaporitic mineral sedimentation. Sedimentation is controlled by the density of Lake water - which is a temperature governed parameter (greatest density at + 4<sup>0</sup> C), the concentration of dissolved salts / solids, and the amount of suspended sediment.

The sedimentation within the lakes of Schirmacher Oasis ought to have typical characteristics and a number of sedimentological possibilities may exist due to the density differences within the water mass at various point of time. During the Austral Summer, warm water is developed at the surface of the lake, and cool water at the base; whereas cold water is formed at the surface during the winter period. Such a pattern might result in a complete overturning of the water column.

The response of lacustrine systems to simultaneous climatic, tectonic and morphological forcing is poorly understood. Sedimentary records of lakes provide a complete and fairly high resolution data because climate is one of the key factors of lacustrine evolution. Smaller and geologically recent lacustrine systems like that of Schirmacher Oasis are very significant as they are very sensitive to even minor tectonic or climatic changes.

Diatoms and unicellular algae with shells of silica that remains in the sediments make excellent bioindicators. Diatom community composition changes with environmental changes in acidity, nutrient availability and lake circulation. By examining relationships between the modern diatom

communities, their environment, and the diatom communities occurring in the lake sediment cores at various depths, various environmental changes can be reconstructed quantitatively. Similarly, palynological studies constitute very significant aspect in palaeoclimatic studies. Mineral magnetic susceptible measurements have also been used for palaeoclimatic interpretations, provenance and sediment flux studies. The terrestrial records preserved in the lake bottom sediment cores will be complementary to marine records or to the long ice-core records. Thus, stable isotope studies on Oxygen, Carbon and Nitrogen, study of pollen assemblages, Diatoms, Micro flora and fauna becomes essential in palaeoclimatic investigation besides the conventional sedimentological, geochemical and geomorphic parameters.

A meaningful and significant climate change study utilising Antarctic lake bottom sediment cores will require a multi-disciplinary approach in which collaborative studies having specializations in the fields of Palynology, Mineral Magnetic susceptible studies, micro-palaeontological investigations, chronological techniques (AMS / radio carbon dating) besides geochemical and sedimentological inputs is essential.

#### **Sediment Coring and the Location Details**

A total cumulative length of Lake Bottom sediment cores retrieved during the period is 43. 50 m. The coring device for retrieval was also successfully used during the previous expedition (Jayapaul D, et al., 2010). Various components and some of the cores retrieved from the lake bottoms are shown in **Fig. 1**(a,b,c,d,e and f) whereas location details and the spatial distribution of lakes from where cores were raised have been shown in **Fig. 2**.

**Table 1—Details of lake sediment cores retrieved during 24<sup>th</sup> IAE**

| Sl. No. | Date of Coring | Coordinates                         | Core Length | Number of the Lake |
|---------|----------------|-------------------------------------|-------------|--------------------|
| 1       | 13/03/05       | 70° 45' 59.08" S – 11° 44' 23.04" E | 50 Cm       | L – 49             |
| 2       | 13/03/05       | 70° 45' 58.00" S – 11° 44' 26.00" E | 100 Cm      | L – 49             |
| 3       | 14/03/05       | 70° 45' 57.09" S – 11° 44' 27.00" E | 140 Cm      | L – 49             |
| 4       | 17/03/05       | 70° 45' 45.08" S – 11° 47' 34.08" E | 53 Cm       | L – 60             |
| 5       | 21/03/05       | 70° 45' 45.09" S – 11° 47' 34.07" E | 80 Cm       | L – 60             |

(Contd.)

**Table 1 (Contd.)**

|    |          |                                       |         |        |
|----|----------|---------------------------------------|---------|--------|
| 6  | 22/03/05 | 70° 45' 46.01" S – 11° 47' 36.08" E   | 93 Cm   | L – 60 |
| 7  | 27/03/05 | 70° 45' 19.03" S – 11° 39' 44. 07" E  | 78 Cm   | L – 27 |
| 8  | 24/03/05 | 70° 45' 19.01" S – 11° 39' 43.09" E   | 62 Cm   | L – 27 |
| 9  | 27/03/05 | 70° 45' 51.06" S – 11° 43' 43.03" E   | 65 Cm   | L – 49 |
| 10 | 28/03/05 | 70° 45' 59.03" S – 11° 44' 27.01" E   | 100Cm   | L – 49 |
| 11 | 28/03/05 | 70° 45' 48.07" S – 11° 44' 17. 02" E  | 150Cm   | L – 49 |
| 12 | 02/04/05 | 70° 45' 21.05" S – 11° 35'. 51.02" E  | 126Cm   | L – 6  |
| 13 | 02/04/05 | 70° 45' 21.05" S – 11° 35' 51. 06" E  | 124Cm   | L – 6  |
| 14 | 03/04/05 | 70° 45' 12.01" S – 11° 40' 13. 08" E  | 35Cm    | L – 27 |
| 15 | 03/04/05 | 70° 45' 16' 04" S – 11° 39' 53. 06" E | 57Cm    | L – 27 |
| 16 | 03/04/05 | 70° 45' 19.02" S – 11° 39' 44. 01" E  | 53Cm    | L – 27 |
| 17 | 03/04/05 | 70° 45' 45.05" S – 11° 40' 43. 05" E  | 35Cm    | L – 33 |
| 18 | 09/04/05 | 70° 46' 25.05" S – 11° 49' 41. 02" E  | 39 Cm   | L – 33 |
| 19 | 27/04/05 | 70° 45' 51.06" S – 11° 44' 29. 00" E  | 47Cm    | L – 49 |
| 20 | 27/04/09 | 70° 45' 48.08" S – 11° 44' 28. 02" E  | 194Cm   | L – 49 |
| 21 | 12/04/05 | 70° 46' 25.25" S – 11° 49' 41. 02" E  | 46 Cm   | L - 75 |
| 22 | 9/09/05  | 70° 45' 52.00" S – 11° 43' 43. 04" E  | 66 Cm   | L – 49 |
| 23 | 28/10/05 | 70° 45' 57.00" S – 11° 44' 25. 00" E  | 90Cm    | L – 49 |
| 24 | 6/11/05  | 70° 45' 21.03" S – 11° 35' 51. 05" E  | 150Cm   | L-6    |
| 25 | 8/11/05  | 70° 45' 21.42" S – 11° 35' 51. 09" E  | 173Cm   | L-6    |
| 26 | 8/11/05  | 70° 45' 21.42" S – 11° 35' 51. 02" E  | 162Cm   | L-6    |
| 27 | 15/11/05 | 70° 45' 48.09" S – 11° 44' 28. 06" E  | 149 Cm  | L – 49 |
| 28 | 16/11/05 | 70° 45' 47.04" S – 11° 44' 07. 02" E  | 35 Cm   | L – 49 |
| 29 | 19/11/05 | 70° 45' 48.09" S – 11° 44' 28. 06" E  | 221Cm   | L – 49 |
| 30 | 19/11/05 | 70° 45' 48.09" S – 11° 44' 28. 06" E  | 198Cm   | L – 49 |
| 31 | 21/11/05 | 70° 45' 48.9" S – 11° 44' 28.4" E     | 204.5Cm | L – 49 |
| 32 | 21/11/05 | 70° 45' 49.00" S – 11° 44' 28.06" E   | 197Cm   | L – 49 |
| 33 | 23/11/05 | 70° 45' 49.00" S – 11° 44' 28.06" E   | 204.5Cm | L – 49 |
| 34 | 23/11/05 | 70° 45' 49.00" S – 11° 44' 28.06" E   | 186Cm   | L – 49 |
| 35 | 24/11/05 | 70° 45' 49.1" S – 11° 44' 33.5" E     | 138Cm   | L – 49 |
| 36 | 25/11/05 | 70° 45' 51.5" S – 11° 43' 41. 8" E    | 118Cm   | L – 49 |
| 37 | 25/11/05 | 70° 45' 51.5" S – 11° 43' 41. 8" E    | 118Cm   | L – 49 |
| 38 | 27/11/05 | 70° 45' 21.4" S – 11° 35' 52. 5" E    | 97 Cm   | L-6    |
| 39 | 27/11/05 | 70° 45' 21.7" S – 11° 35' 52. 8" E    | 124 Cm  | L-6    |



Fig. 1 : (A, B, C)-Indigenously designed components of coring device, the core catcher and corer fitted with sample tube and the outer steel tube (D)-Inserted coring device into frozen lake for in-situ sediment core retrieval. (E)-The retrieved sediment core (F)-Frozen sediment cores for safe transportation of samples for multi-disciplinary paleo-climatic studies

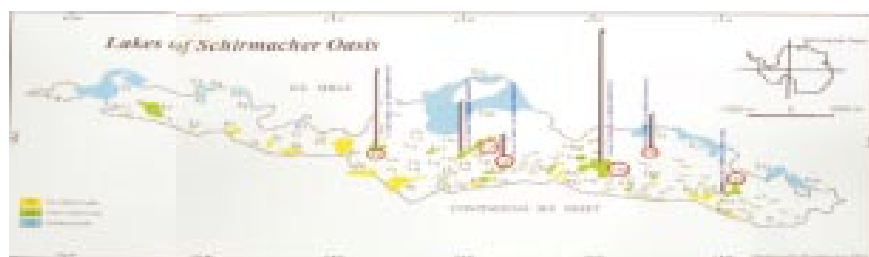


Fig. 2 : The distribution of lakes in Schirmacher Oasis and the lakes from where sediment cores have been retrieved

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